XRPL Validator — Daily Engineering Report · 2026-08-10

Period: 2026-08-10 00:00 to 24:00 US Eastern (2026-08-10 04:00 to 2026-08-11 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(xrpl-ledger): reprice bridged leg A after the mid-leg rounds up** (`07f2e32`, xrpl-validator) 9bb5ceb rounded the bridged XRP mid-leg UP to whole drops and left `gets_in` priced from the FRACTIONAL value it had before. So leg A took a whole drop and paid for a fraction less of one, every time the rounding did anything.
- **fix(xrpl-ledger): a payment pays the INPUT issuer's TransferRate** (`c99ebb2`, xrpl-validator) `apply_path_payment` charged the OUTPUT issuer's rate (`want_rate`) and let the input through free. A payment sets `ownerPaysTransferFee_ = false`, so `trIn = transferRate(book_.in.account)`: the SENDER parts with `amount × rate` and the counterparty receives `amount` (BookStep.cpp:737-739). `apply_iou_direct` has always done this for the same-currency case — the crossing path never did.
- **fix(xrpl-ledger): match rippled's AMM slice sizing on direct offer crossing** (`b2c47da`, xrpl-validator) Two coupled rules, both read off libxrpl's own trace of #106143011 4607B92B against the RLUSD/XRP pool and confirmed in rippled's source.
- **fix(xrpl-node): choose the hydration RPC per LEDGER, not per sweep script** (`25c9e31`, xrpl-validator) The gate corpora AGE OUT of our LAN validator's rolling window (~290k ledgers on 2026-08-10, 105919185-106209950), so a fixture that hydrated locally last month must come from s2 today. A server pinned per script cannot express that, and both failure directions were live: * `batch2sweep.sh` pins `--rpc <lan>` while 7 of its 26 ledgers had rolled out of the window (105909285 105909624 105910649 105911834 105912291 …
- **fix(xrpl-ledger): mint real LP tokens for AMMCreate and single-asset deposits** (`85bb7df`, xrpl-validator) Both mint sites were hardcoded 1e7 placeholders, documented as "the magnitude is oracle-corrected downstream". They were 23 of the 83 value divergences at gate 46 and by far the largest in the backlog — relative errors up to 3e5, where the whole remaining OfferCreate tail sits below 2e-4.
- **fix(xrpl-ledger): quantize a deposit's LP mint to the pool balance's ulp** (`acf49ab`, xrpl-validator) SETTLES THE MANTISSA-SCALE QUESTION, which `85bb7df` left open as the suspected cause of the single-asset deposit precision tails. It is NOT the cause.

Independence program

No independence record for UTC day 2026-08-10; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **MPC Ceremony Attestation on SignerListSet** (custody) Add SFields sfMPCCeremonyHash (Hash256) and sfMPCQuorum (t-of-n) plus a tfMPCAttested flag to SignerListSet; when set, the transactor requires the has
- **Vault NAV Marking for Tokenized Funds** (rwa) Add sfNAVPerShare (Number) and sfNAVManager (AccountID) SFields to the SingleAssetVault ledger object, and a new VaultMarkNAV transaction, signable on
- **Iceberg Offers with TickSize-Quantized Clips** (dex) Add a tfIceberg flag to OfferCreate plus an sfDisplayQuantity SField (the visible clip, required to be a multiple of the trading pair's TickSize preci

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
08-10 05:37	—	xrpl-validator	`07f2e32`	fix(xrpl-ledger): reprice bridged leg A after the mid-leg rounds up
08-10 14:25	—	xrpl-validator	`c99ebb2`	fix(xrpl-ledger): a payment pays the INPUT issuer's TransferRate

Committed	Merged into main	Repo	Commit	Subject
08-10 17:03	—	xrpl-validator	`b2c47da`	fix(xrpl-ledger): match rippled's AMM slice sizing on direct offer crossing
08-11 00:36	—	xrpl-validator	`25c9e31`	fix(xrpl-node): choose the hydration RPC per LEDGER, not per sweep script
08-11 02:24	—	xrpl-validator	`85bb7df`	fix(xrpl-ledger): mint real LP tokens for AMMCreate and single-asset deposits
08-11 03:23	—	xrpl-validator	`acf49ab`	fix(xrpl-ledger): quantize a deposit's LP mint to the pool balance's ulp

Times in UTC. A commit appears on the day it was committed to main or merged into it.